

Tutorial 3

Part 1

Three counts (numbers of individuals per 10 m²) in four seasons for two different species of stick insect were determined with the results in Table 1.21. Analyse and interpret these data.

Season	Megacrania sp	Extatosoma sp	Total
Spring	15	13	
	18 (55)	12 (36)	
	22	11	91
Summer	12	15	
	14 (42)	19 (55)	
	16	21	97
Autumn	10	23	
	11 (30)	25 (74)	
	9	26	104
Winter	6	30	
	8 (21)	29 (90)	
	7	31	111
Total	148	255	403

Note: Numbers in parentheses are subtotals for each season and species.

1. What design has been used here and what is the treatment structure? Be specific in naming the design.
2. Why is this not a randomized complete block design?
3. Identify the following based on the experiment description:
 - a. Treatment factor(s):
 - b. Factor levels:
 - c. Treatments:
 - d. Response:
 - e. Experimental units:
 - f. Number of replicates:
4. How was randomization done?
5. Write down the population model equation used for analysing data from this type of experiment. Be specific.
6. What are the assumptions of this model?
7. Calculate (without using R) and define the following:
 - a. The treatment means.
 - b. Overall mean
 - c. Main effects
 - d. Interaction effects

8. Write out the model equation for observations belonging to one of the treatments (you can choose which treatment).
9. What is the standard error for treatment means? For main effects?
10. Construct an interaction plot and identify what effects are present.
11. What are the hypotheses for this ANOVA test?
12. Conduct the ANOVA hypothesis test by hand, present the results in a table.
13. How would you interpret each of the mean squares?
14. What conclusion can you draw from the table about the effects? Calculate p-values using tables and R.
15. Fit the model in R, extract the parameter estimates and obtain the ANOVA table. The data is in a CSV file called “stick_insects.csv”.

Part 2

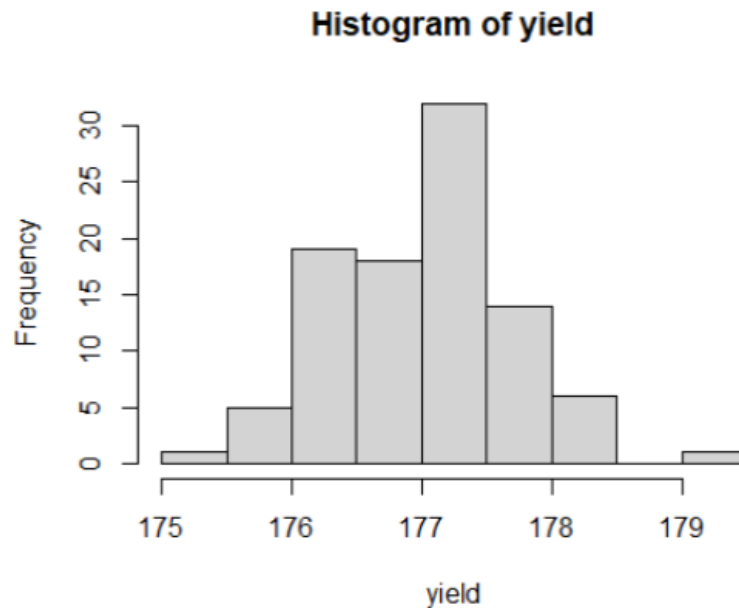
Complete an R analysis with the data below:

Here we are concerned with possible differences in yield among types of seed, among types of fertilizer, and the possibility that the two factors may interact. The data is in a CSV file called “maize_yield.csv”.

Part 3

1. You are researching which type of fertilizer and planting density produces the greatest crop yield in a field experiment. You assign different plots in a field to a combination of fertilizer type (1, 2, or 3) and planting density (1 = low density, 2 = high density), and measure the final crop yield in bushels per acre at harvest time. To start off you perform a two-way ANOVA to test if there are significant differences in crop yield means for the given factors.

Source	Df	Sums of squares	Mean squares	F
fertilizer	A	6.07	3.03	D
density	B	5.12	5.12	15.19
fertilizer: density	C	E	0.21	0.63
Residuals	90	30.34	0.34	



- a) Determine the values of A, B, C, D, and E.
 - b) State the hypotheses associated with the third line in the ANOVA output. What conclusion is reached?
 - c) Comment on the ANOVA assumptions.
 - d) How many observations were used in this experiment?
2. You have recently started a new business and are interested in advertising in your local newspaper. You find data that gives you the number of inquiries resulting from an advertisement. The data also provides information on which day of the week (Monday to Friday) and section of the paper (news, business, or sports) the advertisement was posted. You perform a two-way ANOVA to investigate.

Source	Df	Sum Sq	Mean Sq	F
Section	2	53.73	26.87	15.30
Day	4	146.83	36.71	20.91
Section:Day	8	135.77	16.97	9.67
Residuals	45	79.00	1.76	

Is there reason to believe that you should consider in which section and on which day you should post your advertisement? You should mention the different variables considered, and associated statistics.

3. List all the ANOVA assumptions and suggest a way of checking if each of the assumptions holds true.

4. You are researching if task performed on a computer (point-select, drag-select, etc.) and/or the device used (mouse, stylus, etc.) affects the time it takes to complete a task. The data observed is shown below.

	Device 1	Device 2	Device 3
Task 1	11, 10, 10	15, 17, 13	20, 11, 20
Task 2	18, 14, 23	13, 15, 20	14, 13, 16

At the 10% significance level test whether there exists a significant interaction between Task performed and Device used.

5. In a study on the effect of advertising, TV ads were shown to a group of twenty subjects. Afterwards their attitude to the advertised product was measured. Of the twenty, ten had been classified by a psychologist as having an extrovert personality, while the other ten had an introvert personality. Five subjects from each personality group were randomly chosen to hear the advertisement at high volume and the other five heard it at normal volume.

- a) Calculate as many as possible of the missing values in the ANOVA table below.

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Personality	1				
Volume					4.61
Interaction			15		11.39
Residuals					

- b) Is there evidence that the mean for high volume differs from that for normal volume? Give a p-value as far as the tables will allow.
- c) Do the factors Personality and Volume interact? Test using a p-value approach.
- d) Given the following means:

	High volume	Low volume
Extrovert	10.6	7
Introvert	7.2	6

Draw a sketch to show the nature of the interaction, if any. Then, interpret this in words.

6. In a medical study, the growth of bacteria is thought to be affected by temperature and pH. The experimental data below are of bacterial growth at four different temperatures and three different pH levels. There are two replications. The design is thus a 4×3 factorial design with two replications.

Temperature (°C)	pH 5.5	pH 6.5	pH 7.5
25 rep 1	9	18	36
25 rep 2	11	20	44
30 rep 1	13	23	27
30 rep 2	17	27	33
35 rep 1	18	27	23
35 rep 2	22	33	27
40 rep 1	22	20	7
40 rep 2	28	24	13

Given that the sums of squares for the interaction is equal to 1203, carry out an ANOVA to investigate the effects of temperature and pH on bacterial growth. Is there evidence that the two factors interact?

7. “If the investigator confines his attention to any single factor we may infer either that he is the unfortunate victim of a doctrinaire theory as to how experimentation should proceed, or that the time, material or equipment at his disposal is too limited to allow him to give attention to more than one aspect of his problem...”

This is a quote from R. A. Fisher. What is he referring to?

8. Researchers at a trauma centre wished to develop a program to help brain-damaged trauma victims regain an acceptable level of independence. An experiment involving 72 subjects with the same degree of brain damage was conducted. The objective was to compare different combinations of psychiatric treatment and physical therapy. Each subject was assigned to one combination of four types of psychiatric treatment and six physical therapy programs. There were three subjects in each combination. The response variable is the number of months elapsing between initiation of therapy and time at which the patient was able to function independently.

Construct a skeleton ANOVA (sources of variation and degrees of freedom only) for this experiment.

9. Here are the results of a factorial study involving two treatment factors A and B. Given that SS Interaction is equal to 80, carry out an Analysis of Variance and give clear conclusions to the three hypothesis tests performed.'

	B1	B2	B3
A1	5	9	3
	3	9	8
	3	13	3
	8	6	3
	6	8	3
A2	0	0	0
	2	0	3
	0	0	7
	0	5	5
	3	0	5